Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

Department of Information and Technology

Final Year Project Presentation

on

Signal Buddy : AI Powered Adaptive Traffic Signal Management

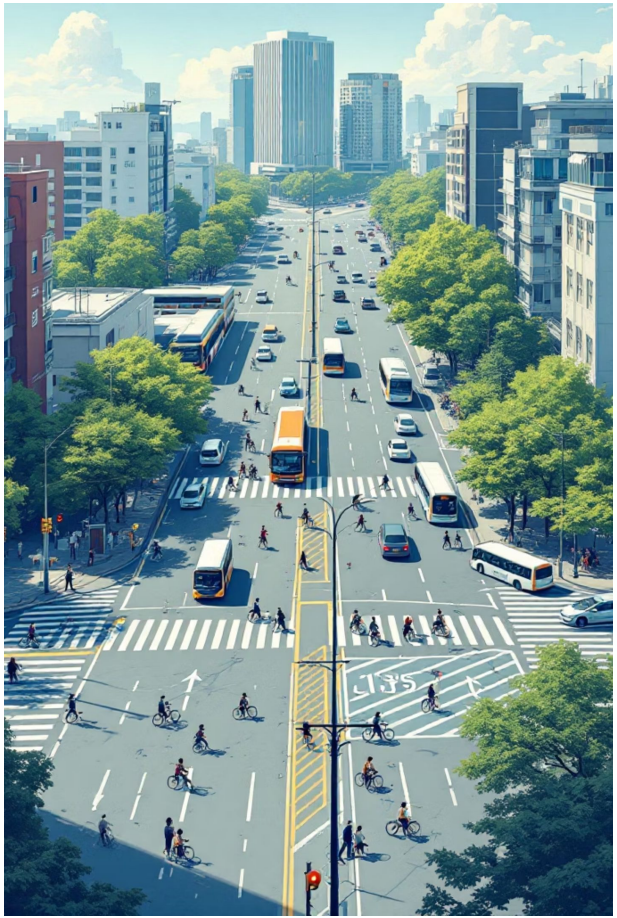
Submitted to: Ms. Shalini Singhal
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Session:
2025-26

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Signal Buddy

AI-Powered Adaptive Traffic Signal Management



Swami Keshvanand Institute of Technology, Information Technology | 2025-2026

Signal Buddy : Adaptive Traffic Signal Management

The Urban Traffic Crisis in Jaipur

30-40%

Wasted Green Time

Static signals waste 30-40% of green time during off-peak hours

15-25

Peak Hour Delays

Average waiting time per intersection during rush hours

₹500Cr

Annual Economic Loss

Lost productivity and fuel costs across Jaipur

50K

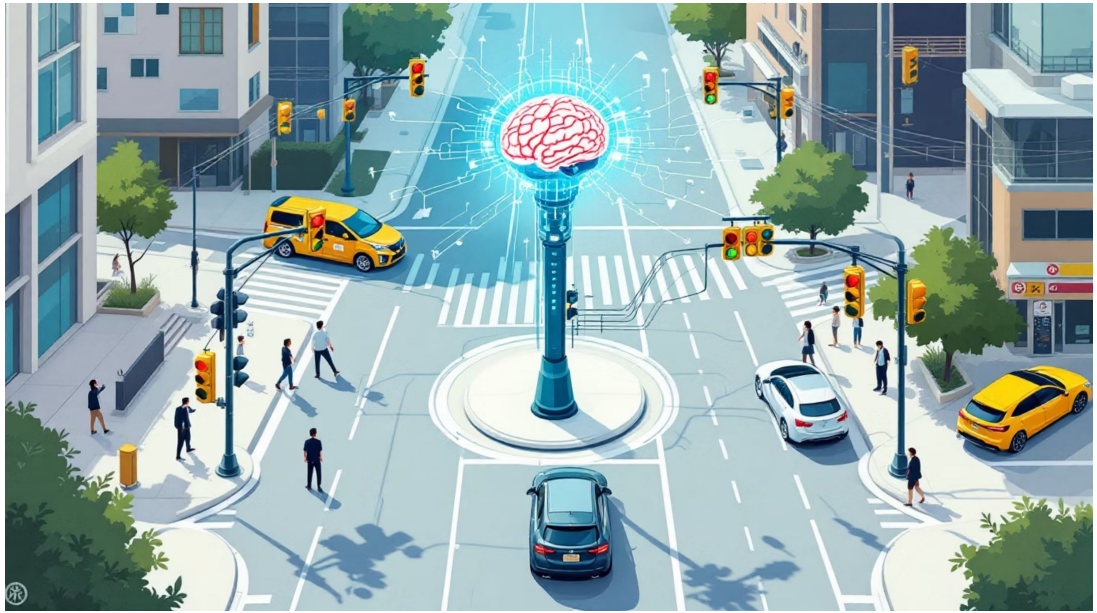
Tons of CO2

Excess emissions from idling vehicles yearly

Traditional traffic signals operate on fixed timers, ignoring real-time traffic flow. This inefficiency causes cascading delays, increased fuel consumption, and frustrated commuters throughout Jaipur's busiest corridors.



Our Solution: The AI Brain for Traffic Lights



Real-Time Intelligence

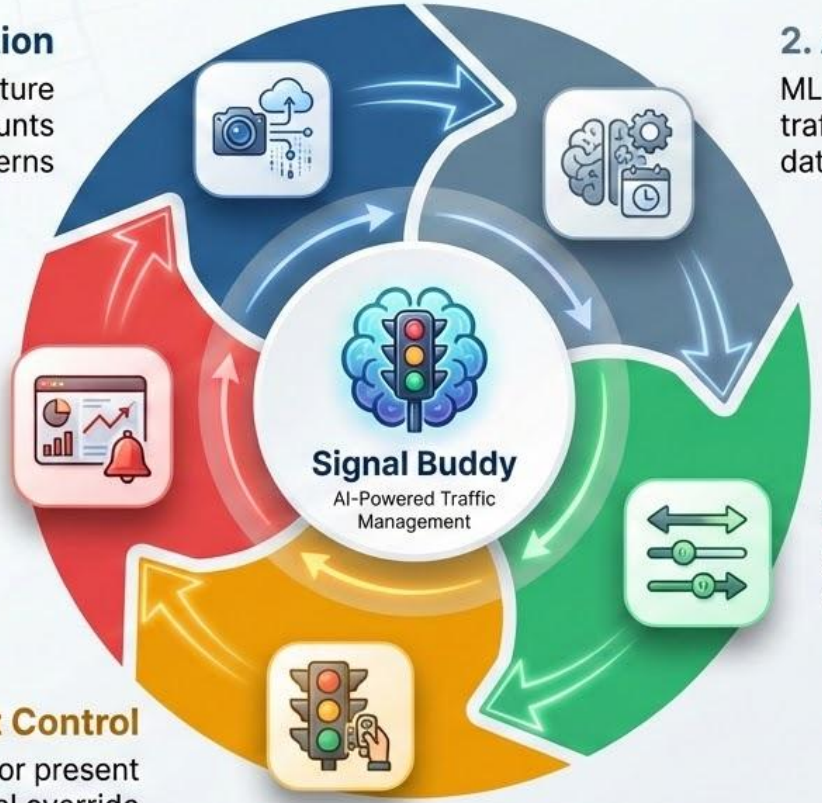
Signal Buddy transforms static traffic lights into adaptive, learning systems that respond to actual road conditions.

- Continuous queue monitoring across 4 directions
- Machine learning predictions for next 5-minute flow
- Dynamic signal timing adjustments
- Operator dashboard with manual override

How Signal Buddy Works

1. Live Data Collection
Cameras and APIs capture real-time vehicle counts and traffic patterns

2. AI Prediction Engine
ML models forecast short-term traffic flow using historical data and weather



3. Adaptive Optimization
System calculates optimal green splits and priority directions

4. Smart Control
Signals auto-adjust or present recommendations for manual override

5. Monitor & Alert
Dashboard displays status and sends notifications for anomalies



Technology Stack

Frontend

- React + TypeScript
- Tailwind CSS
- Leaflet Maps
- Recharts

Backend

- FastAPI
- Python 3.11
- PostgreSQL
- Redis

ML & Data

- scikit-learn
- ARIMA
- Prophet
- OpenCV

Additional integrations include Google Maps API for traffic data and OpenWeather API for environmental factors. The system connects to existing traffic signal hardware through standard control interfaces.

Live Dashboard Interface

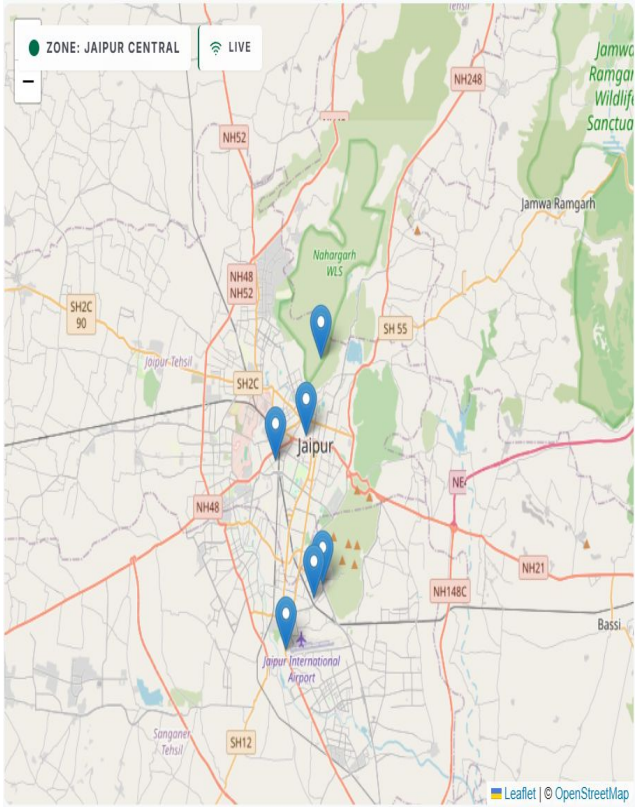
Real-Time Monitoring

- Live map showing 4 intersections
- Queue status per direction
- Signal countdown displays

AI-Powered Insights

- Predicted flow recommendations
- Emergency all-red capability
- Manual override controls

Simulation Result: 87% congestion reduction achieved in controlled testing environment



Made with GAMMA

Key Feature

A. RBAC Security

- **Admin:** Full system oversight & analytics
- **Operator:** Incident reporting & overrides
- **Viewer:** Read-only monitoring

B. Incident Management

- **Categorization:** Severity tagging (Low to Critical)
- **Lifecycle:** Reported → In Progress → Resolved
- Full audit trail for city management

C. AI Decision Support

- Analyzes queue counts for signal adjustments
- Suggests: "Extend Green by 15s"
- **Human-in-the-loop:** Operator approval required

D. Data & Analytics

- **CSV Exports:** Raw processing in BI tools
- **PDF Reports:** Professional trend summaries
- Visual incident frequency mapping

ML Prediction Engine

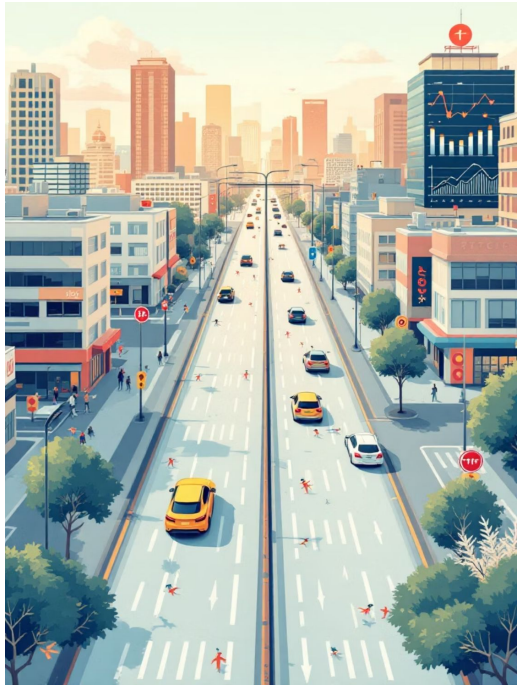
Traffic Flow Prediction

Input Data: Historical counts, weather conditions, time of day, day of week

Model: Hybrid approach using ARIMA for seasonality and Prophet for trend analysis

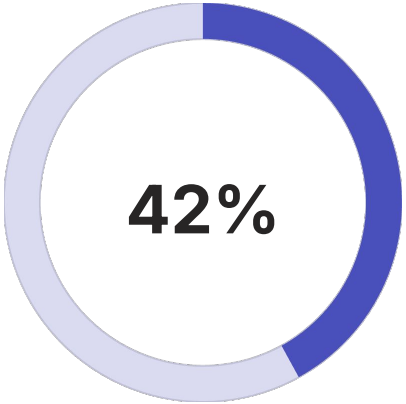
Accuracy: 87% on validation dataset with 10-second refresh rate

```
def predict_traffic(queue_length, weather, hour):
    # ARIMA handles historical patterns
    # Prophet adjusts for external factors
    # Returns predicted flow percentage
    return forecast
```

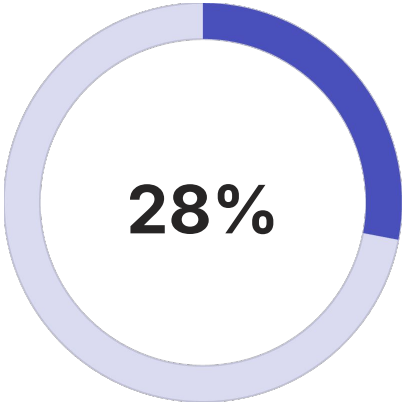


Impact & Results

One-Week Simulation at 4 Jaipur Intersections



Delay reduction from 4.1 to 2.4 minutes average



Throughput increase in vehicles per hour

Economic Benefits

- ₹45,000 monthly fuel savings
- 50K+ hours productivity recovered
- Quick ROI on implementation

Environmental Impact

- 120 kg daily CO2 reduction
- Lower emissions from reduced idling
- Sustainable urban mobility

System Architecture & Roadmap

Current: Phase 1

Single intersection control with AI prediction and manual override

1

2

Phase 2: Coordination

Multi-intersection synchronization and YOLO vehicle counting from cameras

Phase 3: Mobile

Field officer app with real-time alerts and municipal signal integration

3

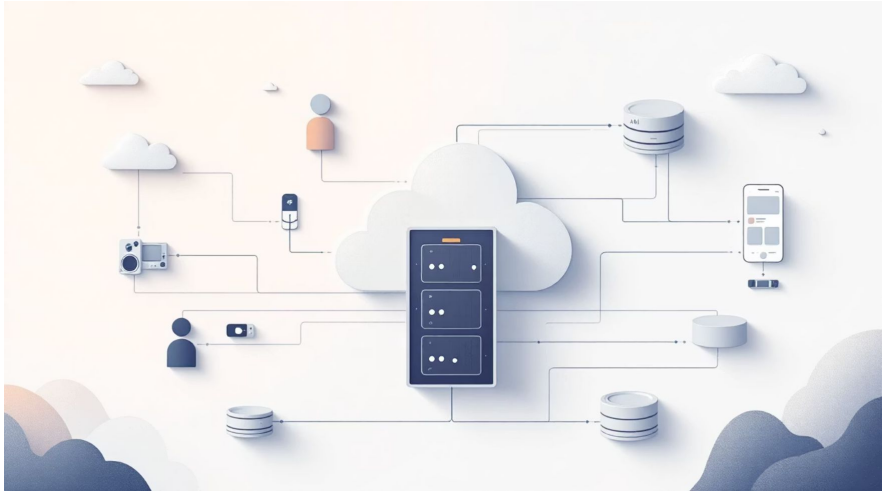
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Phase 4: Advanced

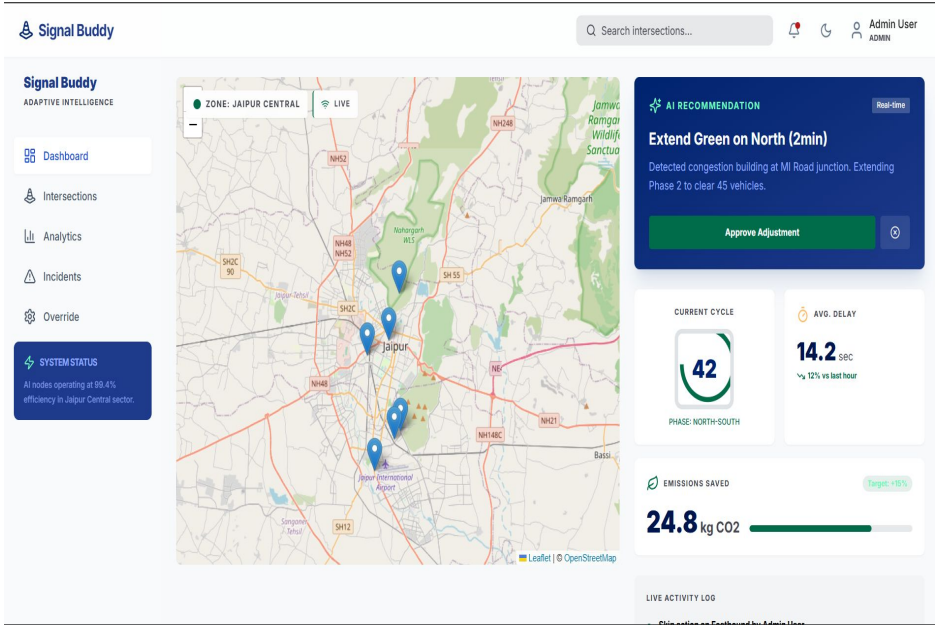
Blockchain audit trail and full municipal deployment

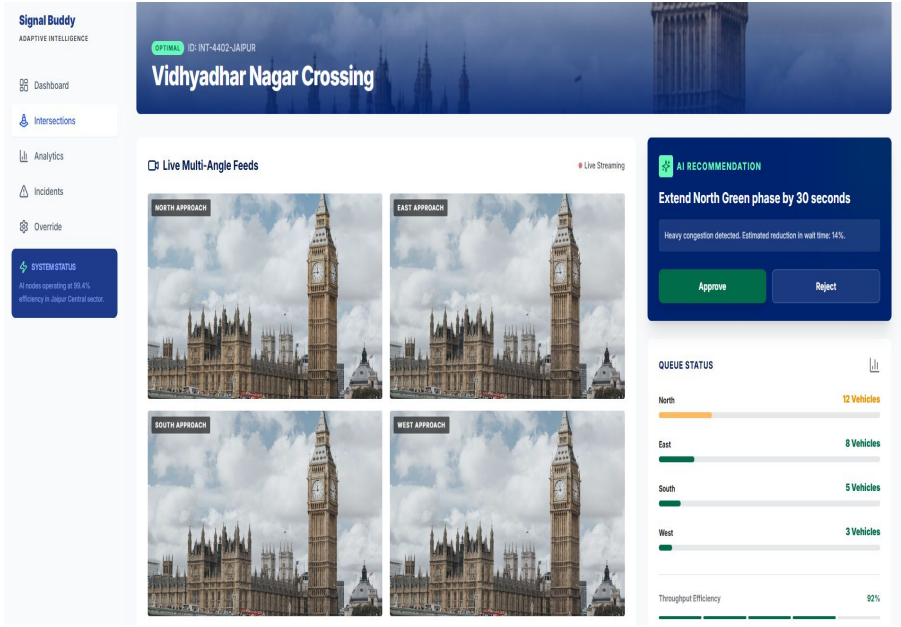
End-to-End Flow

Cameras and APIs feed data to FastAPI backend. ML models process predictions stored in PostgreSQL. React dashboard displays live maps while Redis caches frequent queries for hardware signal control.

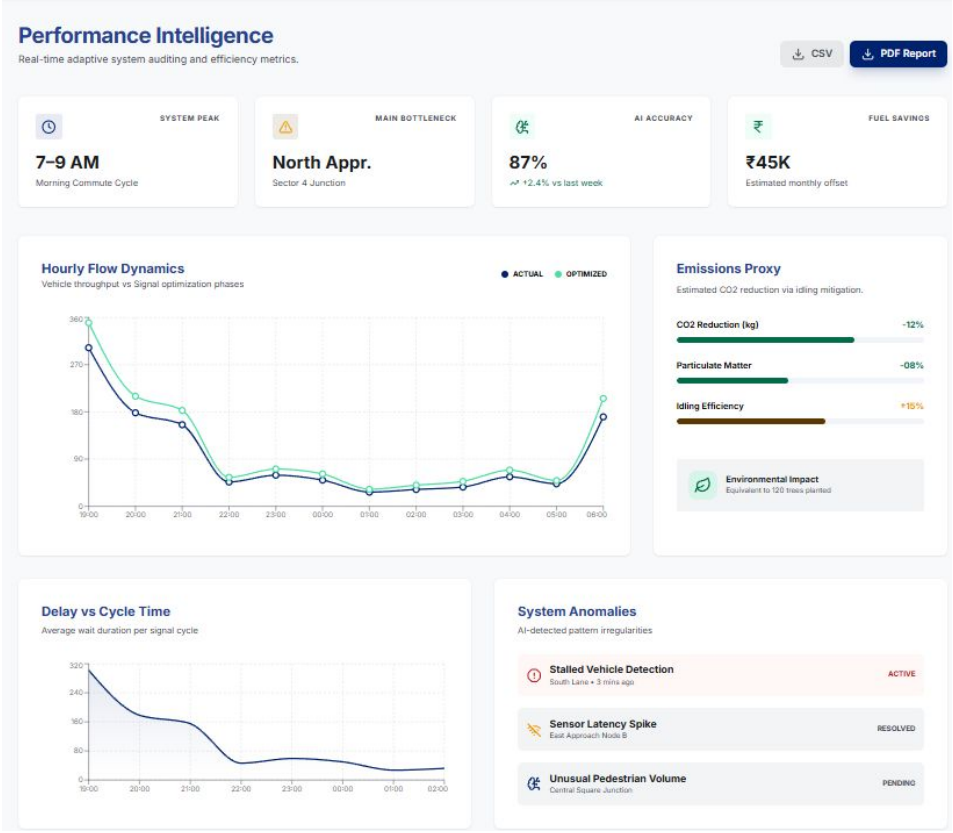


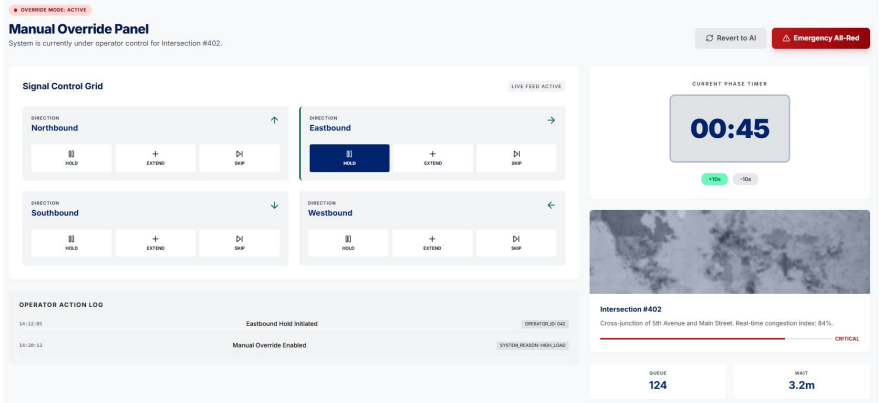
Project Snippets:

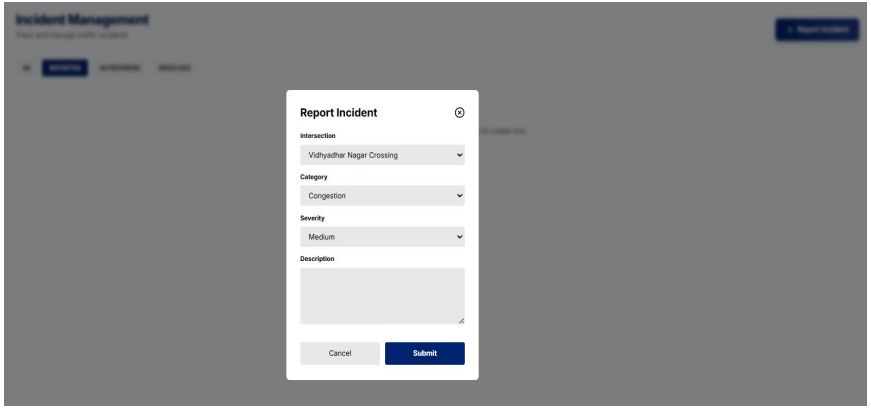




Project Snippets:







References



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THANK YOU